

SWASTHYASYNC DIGITAL HEALTH PLATFORM

Comprehensive Technical Stack, System Architecture & Engineering Specifications

Version: 2.4.0 (Pan-India Production Edition) | **Target:** All 36 Indian States & UTs | **Compliance:** DPDP Act 2023 & ABDM/EHR Standards

1. Architecture Overview & Design Principles

SwasthyaSync is engineered as a zero-friction, location-aware, reactive single-page architecture built with modern Web Standards. It delivers clinical responsiveness across varying connectivity environments (urban teaching hospitals to rural Primary Health Centres) without hardcoding regional assumptions.

Layer / Tier	Technologies & Libraries	Architectural Role & Functionality
Frontend Framework	React 18.3+ / TypeScript 5.5	Component-driven reactive UI, strictly typed interfaces, strict null checks, dynamic role branching.
Build & Tooling	Vite 6.4 (ESBuild Engine)	Ultra-fast sub-second Hot Module Replacement (HMR), tree-shaking, optimized ES module chunking.
Design System & UI	Tailwind CSS 3.4 / Lucide React	Utility-first design tokens, modern responsive cards, accessible modal dialogs, 50+ SVG medical icons.
Geographic & Maps	Leaflet 1.9 / OpenStreetMap API	Pan-India interactive mapping, live Overpass facility discovery, Haversine geospatial proximity engine.
State & Context	React Context + Pub/Sub Services	Centralized UserLocationContext, AuthContext, ToastContext with zero-dependency pub/sub streaming.
Localization (i18n)	Custom Multi-Dictionary Engine	Instant client-side translation across 23 official Indian languages with RTL support for Urdu.
Backend & Auth	Supabase (PostgreSQL + GoTrue)	PostgreSQL relational store, Row-Level Security (RLS), JWT token sessions, prototype offline persistence.
Security & Compliance	DPDP Act 2023 & ABDM/EHR	Encrypted localStorage keys, granular RBAC (Patient, Doctor, Admin), data privacy safeguards.

2. Core Subsystems & Technical Details

A. Centralized Location-Aware Identity System

- **Unified Hierarchy:** Country (India) → State/UT (36) → District (700+) → Sub-District/Taluka → Healthcare Facility.
- **Contextual Isolation:** Doctors and Administrators inherit facility and jurisdictional scopes directly from their profile without global state leakage.
- **Live Location Sync:** Seamless fallback from browser GPS coordinates to registered domicile address.

B. Comprehensive 23-Language Localization Engine

- **Supported Languages:** Assamese, Bengali, Bodo, Dogri, English, Gujarati, Hindi, Kannada, Kashmiri, Konkani, Maithili, Malayalam, Manipuri, Marathi, Nepali, Odia, Punjabi, Sanskrit, Santali, Sindhi, Tamil, Telugu, and Urdu.
- **Architectural Rule:** Single global header dropdown controller with persistent localStorage memory and native script rendering.
- **Bidirectional Layout:** Automatic dynamic RTL document alignment for Urdu (``dir='rtl'``).

C. Real Hospital Discovery & Leaflet Geospatial Engine

- **Dynamic OSM Query:** Queries OpenStreetMap Overpass API for real healthcare facilities within bounding radii.
- **Fallback Registry:** Curated pan-India national registry with live bed counters, ICU capacity, and casualty numbers.
- **Spatial Math:** Precision Haversine mathematical distance calculation relative to live GPS coords.

D. AI Report Simplifier & Digital Locker

- **Side-by-Side Comparative UI:** Synchronized split-screen showing raw laboratory document alongside AI plain-language translation.
- **Clinical Parameter Extraction:** Normal vs Abnormal marker badges (CBC, Lipid, HbA1c, Renal Panels).
- **Digital Locker Archival:** Optional-but-recommended ABDM health locker persistence.

3. Deployment & Runtime Requirements

Parameter	Production Specification
Node.js Runtime	v18.x or v20.x+ LTS (Tested on Node.js v24)
Client Browser Support	Chrome 90+, Edge 90+, Firefox 88+, Safari 14+, Mobile Chrome/Safari
Client Bundle Size	~272 kB gzipped (Optimized code-split architecture)
API Protocols	HTTPS / WSS (WebSocket real-time event streaming)
Data Storage	PostgreSQL 15+ / Supabase Auth & Storage / Client IndexedDB Cache